

Institute/Department	UNIVERSITY INSTITUTE OF COMPUTING (UIC)	Program	Master of Computer Applications (MC305)
Master Subject Coordinator Name:	Ajay Yadav	Master Subject Coordinator E-Code:	E13382
Course Name	Python Programming	Course Code	25CAH-606

Lecture	Tutorial	Practical	Self Study	Credit	Subject Type
0	2	4	0	4.00	T

Course Type	Course Category	Mode of Assessment	Mode of Delivery
Major Core	Graded (GR)	Hybrid	Hybrid (HYB)

Mission of the Department	M1. To provide innovative learning centric facilities and quality-oriented teaching learning process for solving computational problems. M2. To provide a framework through Project Based Learning to support society and industry in promoting a multidisciplinary activity. M3. To develop crystal clear evaluation system and experiential learning mechanism aligned with futuristic technologies and industry. M4. To provide doorway for promoting research, innovation and entrepreneurship skills in collaboration with industry and academia. M5. To undertake societal activities for upliftment of rural/deprived sections of the society.
Vision of the Department	To be a Centre of Excellence for nurturing computer professionals with strong application expertise through experiential learning and research for matching the requirements of industry and society instilling in them the spirit of innovation and entrepreneurship.

Program Educational Objectives(PEOs)

PEO1	Establish a well-fortified computing foundation of successful professionals by applying computing fundamentals and domain-specific knowledge, demonstrating their innovative skills and considering social and environmental concerns.
PEO2	Undertake successful implementation of ethical solutions as an individual or a member or a leader of a team by investigating, analyzing, formulating and solving complex computing problems in multidisciplinary approaches using modern tools.
PEO3	Enhance professionalism and ethical attitude in the profession while communicating with local, national, and foreign peers, bound within regulations and leading to lifelong learning.
PEO4	Promote awareness for uplifting health, safety, legal, environmental, ethical and cultural diversity issues for serving the society.

Program Specific OutComes(PSOs)

PSO1	Analyze their abilities in systematic planning, developing, testing and executing complex computing applications in field of Social Media and Analytics, Web Application Development and Data Interpretations.
PSO2	Apprise in-depth expertise and sustainable learning that contributes to multi-disciplinary creativity, permutation, modernization and study to address global interest.

Program OutComes(POs)

PO1	Apply mathematics and computing fundamental and domain concepts to find out the solution of defined problems and requirements.
PO2	Use fundamental principle of Mathematics and Computing to identify, formulate research literature for solving complex problems, reaching appropriate solutions.
PO3	Understand to design, analyse and develop solutions and evaluate system components or processes to meet specific need for local, regional and global public health, societal, cultural, and environmental systems.
PO4	Use expertise research-based knowledge and methods including skills for analysis and development of information to reach valid conclusions.
PO5	Adapt, apply appropriate modern computing tools and techniques to solve computing activities keeping in view the limitations
PO6	Exhibiting ethics for regulations, responsibilities and norms in professional computing practices.

PO7	Enlighten knowledge to enhance understanding and building research, strategies in independent learning for continual development as computer applications professional.
PO8	Establishing strategies in developing and implementing ideas in multi- disciplinary environments using computing and management skills as a member or leader in a team.
PO9	Contribute to progressive community and society in comprehending computing activities by writing effective reports, designing documentation, making effective presentation, and understand instructions
PO10	Apply mathematics and computing knowledge to access and solve issues relating to health, safety, societal, environmental, legal, and cultural issues within local, regional and global context.
PO11	Gain confidence for self and continuous learning to improve knowledge and competence as a member or leader of a team.
PO12	Learn to innovate, design and develop solutions for solving real life business problems and addressing business development issues with a passion for quality competency and holistic approach.

Text Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	"Think Python: How to Think Like a Computer Scientist"	Allen B. Downey	2nd edition	Shroff/O'Reilly Publishers	2016
2	MURACH'S PYTHON PROGRAMMING	Michael Urban, Joel Murach, Mike Murach	1st edition	Shroff/Murach	1 January 2017

Reference Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	An Introduction to Python – Revised and updated for Python 3.2	Guido van Rossum and Fred L. Drake Jr	1st edition	Network Theory Ltd.	2011

Course OutCome	
SrNo	OutCome
CO1	Interpret the fundamental Python syntax and semantics and be fluent in the use of Python control flow statements
CO2	Ensure proficiency in the handling of strings and functions
CO3	Determine the methods to create and manipulate Python programs by utilizing the data structures like lists, dictionaries, tuples and sets
CO4	Identify the commonly used conditional statements to control the execution flow
CO5	Construct a program using Object- Oriented Programming concepts such as encapsulation, inheritance and polymorphism as used in Python

Lecture Plan Preview-Theory						
Unit No	LectureNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Numer (s)
1	1	Introduction to python	Introduction to python, features, setting up programming environment	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO1
1	2	Introduction to python	python program structure, Tokens: Keywords, identifier, variables, data types, operators,	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO1
1	3	Introduction to python	Scope and lifetime of variables, Basics of the Python REPL (Read-Eval-Print Loop).	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO1

1	4	Conditional Statements	Concept of indentation If, If Else, Switch; Looping	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4
1	5	Conditional Statements	For, While, Nested loops, Jumping statements, Loop control statements	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4
1	6	Conditional Statements	break and continue.	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4
1	7	Python Sequence	Lists-Accessing elements, Index position, Using individual values from a list, Changing/Adding and removing Elements, Organizing a list	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3

1	8	Python Sequence	Loop through an entire list, Avoiding Indentation Error, Numerical Lists, Sublist, List comprehension, Tuple, Set, Dictionary- Working with dictionaries,	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3
1	9	Python Sequence	Looping through dictionaries, Nesting, Input and output: print() and input() functions, Understanding comments and code structure.	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3
1	10	Function	Passing arguments, Return Values, Passing a list, Passing an Arbitrary number of arguments ,Function in Modules	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO2
1	11	Function	Recursive function, Nested functions, Default and flexible arguments ,Lambda function ,Map()function	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO2

1	12	Function	Filter function ,Reduce()function, Python in built functions, Iterating through data structures	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO2
1	13	Class	Creating and using a class, working with classes and Instances,	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO5
1	14	Class	public and private members, Inheritance, types of inheritance	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO5
1	15	Class	Polymorphism, Importing Classes, Python Standard library, The import statement.	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO5

2	16	Introduction to Data Analytics	Introduction to Data Analytics, requirements of data analytics in Python.	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3
2	17	Introduction to Data Analytics	Introduction to Numpy,	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO2
2	18	Introduction to Data Analytics	features, environment setup,	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO2
2	19	Introduction to Data Analytics	numpy Ndarray, array creation data types	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO2

2	20	Introduction to Data Analytics	array attributes	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO2
2	21	Introduction to Data Analytics	numpy operations	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO2
2	22	Introduction to Data Analytics	mathematical and statistical functions	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO2
2	23	Introduction to matplotlib	Introduction to matplotlib	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3

2	24	Introduction to matplotlib	Figure class, Axes class	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3
2	25	Introduction to matplotlib	line plot, subplots, Bar	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3
2	26	Introduction to matplotlib	plot, histogram, scatter plot, pie chart, box plot	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3
2	27	Introduction to matplotlib	plot, histogram, scatter plot, pie chart, box plot	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3

2	28	Introduction to matplotlib	area chart, word cloud, Bee swarm plot, violin graph	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3
2	29	Introduction to matplotlib	working with text, Customizing	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3
2	30	Introduction to matplotlib	Plots: Titles, labels, and legends.	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO3
3	31	GUI programming using Tkinter	Introduction to Tkinter, Installing Tkinter	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4

3	32	GUI programming using TKinter	Basics of creating a simple	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4
3	33	GUI programming using TKinter	Basics of creating a simple	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4
3	34	GUI programming using TKinter	Tkinter window,Processing events	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4
3	35	GUI programming using TKinter	Tkinter window,Processing events	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4

3	36	GUI programming using TKinter	The widget classes- canvas	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4
3	37	GUI programming using TKinter	geometry managers-display images- menus- popup menus- mouse	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4
3	38	GUI programming using TKinter	geometry managers-display images- menus- popup menus- mouse	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4
3	39	GUI programming using TKinter	geometry managers-display images- menus- popup menus- mouse	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO4

3	40	GUI programming using TKinter	geometry managers-display images- menus- popup menus- mouse	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO5
3	41	GUI programming using TKinter	key events and bindings, Widget properties and methods.	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO5
3	42	Data visualization using seaborn and plotly	Introduction to seaborn, relational plots, categorical plots	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO5
3	43	Data visualization using seaborn and plotly	distribution plots, matrix plots.Plotly:Line, bar	,T-"Think Python: How to Think Li,T-MURACH`S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO5

3	44	Data visualization using seaborn and plotly	scatter, histogram, violin, gantt	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO5
3	45	Data visualization using seaborn and plotly	heatmap, 3D graph.	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flippe d Classes,Info graphics,Inst ructor Lead WorkShop,P PT,Professo r of Practice/Adj unct Faculty/Visiti ng Professor,R eports,Simul ation,Video Lecture	CO5

Lecture Plan Preview-Practical

Unit No	ExperimentNo	Experiment Name	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Numer(s)
1	1	Write a program to print twin primes less than 1000. If two consecutive odd numbers are both prime then they are known as twin primes. Write a program to implement these formulae of permutations and combinations. Number of permutations of n objects taken r at a time: $p(n, r) = n! / (n-r)!$. Number of combinations of n objects taken r at a time is: $c(n, r) = n! / (r!(n-r)!) = p(n, r) / r!$	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographic s,Instructor Lead WorkShop,PPT,Pro fessor of Practice/Adjunct Faculty/Visiting Professor,Reports, Simulation,Video Lecture	CO1

1	2	WAP to perform list operation-creating, accessing, slicing, modifying, adding, removing, sorting, reversing, finding length, and looping through lists. WAP to get user id, user name, and user age from user and based on the entered id print the details for particular user. Hint: use dictionary	,T-"Think Python: How to Think Like a Computer Scientist", T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
1	3	Implement a Student class with information such as rollno, name, class. The information must be entered by the user.	,T-"Think Python: How to Think Like a Computer Scientist", T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO2
1	4	Lab MST-1	,T-"Think Python: How to Think Like a Computer Scientist", T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1,CO2
2	5	1. Write an experiment to swap two columns in numpy array Write an experiment to import a dataset with numbers and texts keeping the text intact in python numpy?	,T-"Think Python: How to Think Like a Computer Scientist", T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3
2	6	Write a python program to generate a simple bar graph using matplotlib. The graph should be properly labeled. Write a python program to generate Pie-chart using matplotlib. The graph should be properly labeled. Write a Python program to plot the function $y = x^2$ using the matplotlib libraries.	,T-"Think Python: How to Think Like a Computer Scientist", T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3

3	7	Create window using Tkinter with two buttons with process control	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO4
3	8	Project Based on the contents of unit-I, II, III	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
3	9	I) Develop project on Library Catalog System	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
3	10	Project on Health Data Analysis	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
3	11	Project on Health Data Analysis	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
3	12	Develop Sales Analysis Dashboard	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5

3	13	Develop Sales Analysis Dashboard	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
3	14	Create a scientific calculator along with proper interface	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO4,CO5
3	15	Create a scientific calculator along with proper interface	,T-"Think Python: How to Think Li,T-MURACH'S PYTHON PROGRAMMING ,R-An Introduction to Python – Re	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO4,CO5

Assessment Model			
Sr No	Assessment Name	Exam Name	Max Marks
1	Hybrid Course All	Practical Evaluations	40
2	Hybrid Course All	End Term Hybrid Theory	60
3	Hybrid Course All	Attendance Marks	2
4	Hybrid Course All	Surprise Test	12
5	Hybrid Course All	Practical MST	10
6	Hybrid Course All	Practical Worksheet/Projects 1	30
7	Hybrid Course All	Practical Worksheet/Projects 2	30
8	Hybrid Course All	Practical Worksheet/Projects 3	30
9	Hybrid Course All	Practical Worksheet/Projects 4	30
10	Hybrid Course All	Practical Worksheet/Projects 5	30
11	Hybrid Course All	Practical Worksheet/Projects 6	30
12	Hybrid Course All	Practical Worksheet/Projects 7	30
13	Hybrid Course All	Practical Worksheet/Projects 8	30

14	Hybrid Course All	Practical Worksheet/Projects 9	30
15	Hybrid Course All	Practical Worksheet/Projects 10	30
16	Hybrid Course All	Quiz	4
17	Hybrid Course All	Assignment/PBL	10
18	Hybrid Course All	MST-1 Hybrid	20
19	Hybrid Course All	MST-2 Hybrid	20